RhodoSim++

User manual

October 2026

Designed by veö, built upon C++ coding work by F. Er & G. Unel. Code translation and further development assisted by Claude.

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1 Introduction

RhodoSim++ is a simulation and design tool for rhodotron electron accelerators that runs entirely in a web browser. It brings together two programs:

- **RhodoSim**, which tracks electrons through the rhodotron's coaxial cavity and its deflecting magnets. RhodoSim++ is compatible with the C++ RhodoSim by F. Er and G. Unel: it reads and writes the same `config.ini` files, and one of its integrators reproduces the C++ code step for step.

- **The Rhodotron Cavity Profiler**, which designs the RF cavity: it solves the cavity's resonant mode for a chosen cross-section and hands the solved field to the tracking.

The interface is available in English and Turkish. Installation does not need any software packages.

If you use this software, please cite: Er, Muhammet Furkan. "Design and simulation software for rhodotron type electron accelerators." MSc thesis, Boğaziçi University, 2024. We acknowledge support from TÜBİTAK under grant 122F304.

2 Getting started

Open `rhodosim.html`[1] in a recent web browser (Chrome, Edge, Firefox or Safari). Everything runs on your computer; the only network access is for the 3D views, which load the three.js library. Your settings are kept in the browser between visits.

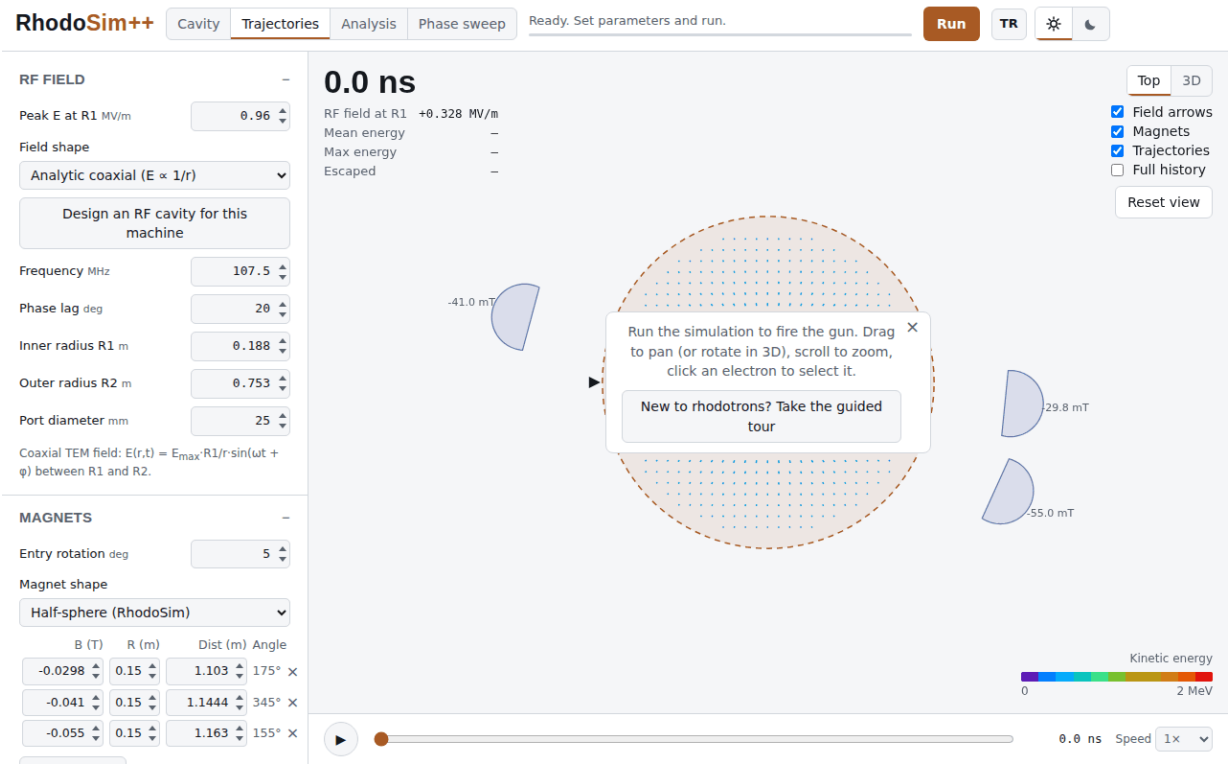


Figure 1: RhodoSim++ when first opened: the header with its four tabs, the settings sidebar on the left, the trajectory view, and the welcome box.

The **header** holds the four tabs (Cavity, Trajectories, Analysis, Phase sweep), a status line, the Run simulation button, the language switch (TR/EN) and the Day/Night theme. The **sidebar** holds the machine's settings and is shared by the three beam tabs; the Cavity tab has its own controls. Below the trajectory view, the **timeline** plays a finished run back.

The welcome box can be closed with its ×; it stays closed. If you are new to rhodotrons, start with the Guided tour (header button, or the welcome box): it introduces the machine step by step, runs small experiments, and restores your own settings when you leave it.

[1] A recent version can be obtained at `https://erkcan.github.io/RhodoSimWeb/Web/rhodosim.html`.

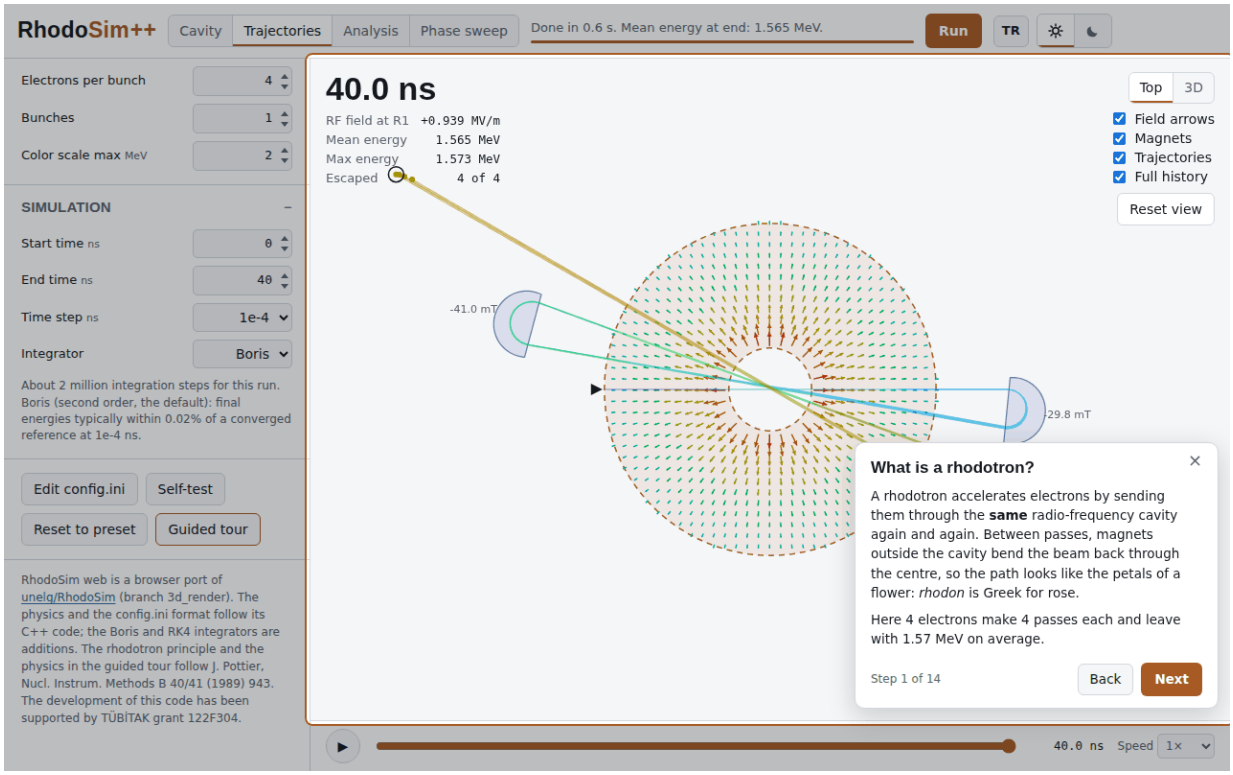


Figure 2: The guided tour. Each step explains one aspect of the rhodotron and shows it in the simulation; some steps invite you to try a change yourself.

3 Trajectories

The Trajectories tab simulates the beam: electrons from the gun cross the cavity, gain energy from its RF field, leave it, are turned around by a magnet, and cross again, pass after pass. Press Run simulation; the status line reports the run's duration and the mean energy at the end.

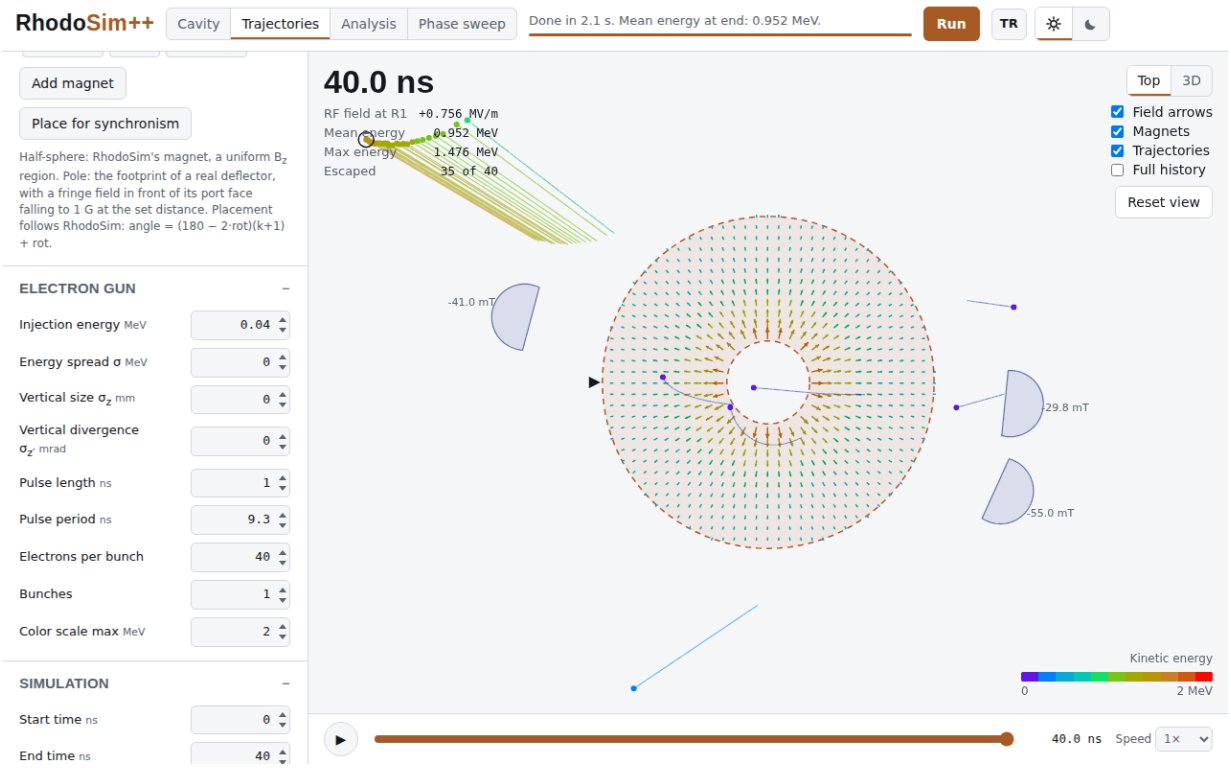


Figure 3: A finished run in the top view. The trajectories are coloured by kinetic energy (the scale is at the bottom right); the readouts at the top left belong to the moment shown on the timeline.

3.1 Settings

RF field. Peak E at R1 (MV/m) is the field amplitude at the inner conductor; Frequency (MHz), Phase lag (degrees), and the conductor radii Inner radius R1 and Outer radius R2 (m) define the cavity. Port diameter (mm) is the opening in the outer conductor through which the beam passes (25 mm for KF-25 ports). Field shape is either Analytic coaxial ($E \propto 1/r$), the field of an ideal coaxial line, or Solved cavity (Cavity tab), the field of a cavity designed in the Cavity tab (Section 7).

Magnets. Each row of the table is one magnet: its field B (T), radius R (m) and distance Dist (m) from the cavity's centre; its angle follows from the Entry rotation. Add magnet appends one. Magnet shape is Half-sphere (RhodoSim), C++ RhodoSim's model (a uniform field in a half-ball), or Pole, fringe field, the footprint of a real deflector with a fringe field in front of its face, whose extent is set by Field 1 G at (m); pole magnets also have a Pole gap (m). Place for synchronism moves each magnet along its line from the centre until the reference electron crosses the centre at the same RF phase on every pass; Undo restores the earlier distances.

Electron gun. Injection energy and Energy spread σ (MeV), Vertical size σ_z (mm) and Vertical divergence $\sigma_{z'}$ (mrad), Pulse length and Pulse period (ns), Electrons per bunch and Bunches. With both vertical spreads at zero every electron stays in the mid-plane; any spread makes the simulation three-dimensional (Section 3.3).

Simulation. Start time and End time (ns), the Time step, and the Integrator: Boris (the default, second order), Runge–Kutta (fourth order) or RhodoSim's own scheme, which reproduces C++ RhodoSim. A note under the integrator estimates the run's cost. Edit config.ini shows the settings as a RhodoSim configuration file, Self-test checks the engine (Section 10), and Reset to preset restores the example machine.

3.2 The view

Top and 3D switch the view. Drag to pan (or rotate in 3D), scroll to zoom, and click an electron to select it; its passes appear in the Analysis tab. The check boxes show or hide Field arrows, Magnets, Trajectories and the Full history; with a solved cavity field, Cavity wireframe draws the cavity's walls. Color scale max (MeV) sets the top of the energy colour scale.

The readouts show the RF field at R1, the Mean energy and Max energy, and how many electrons have Escaped, were Stopped (a numerical breakdown) or were Lost on the machine's apertures. Two labels may appear over the view: Non-physical setup, when a run was made despite a warning (Section 7), and Trajectories from earlier settings: Run to update, when the settings have changed since the run shown. The view always draws the current settings; the trajectories stay those of the last run until you run again.

3.3 Three-dimensional tracking

With a vertical spread at the gun, electrons leave the mid-plane. The RF field and its magnetic field act on them there, pole magnets focus them vertically at their faces, and electrons that rise too far are lost: at a port, when they cross the outer conductor above or below half the Port diameter, or on a pole, above or below half the Pole gap. In the 3D view, the ×1, ×10 and ×50 buttons magnify the beam's heights (millimetres in a metre-sized machine); the cavity stays at true scale.

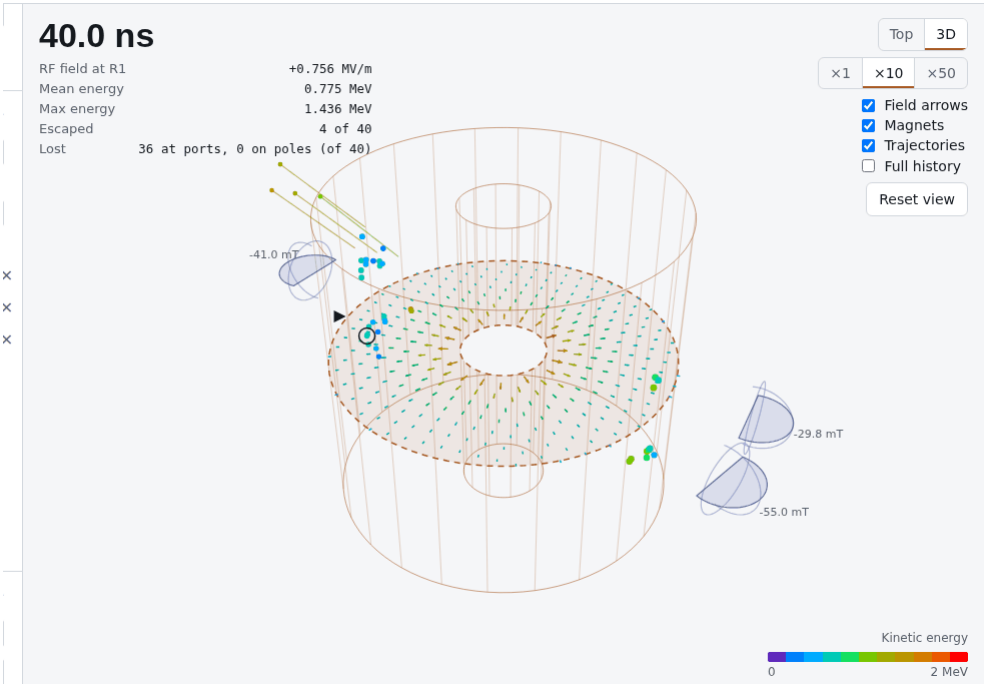


Figure 4: A run with a vertical spread in the 3D view, heights magnified ten times. Lost electrons stay where they hit the outer conductor.

4 Analysis

The Analysis tab studies the last run. Energy over time plots the selected electron's energy, or the bunch's; Energy distribution shows the energies at a chosen moment as a histogram with its statistics. Passes through the cavity lists the selected electron's passes, and All electrons, pass by pass summarises the bunch: for each pass, how many electrons completed it, their energy after it, the RF phase at the centre, and the loop's length in RF periods.

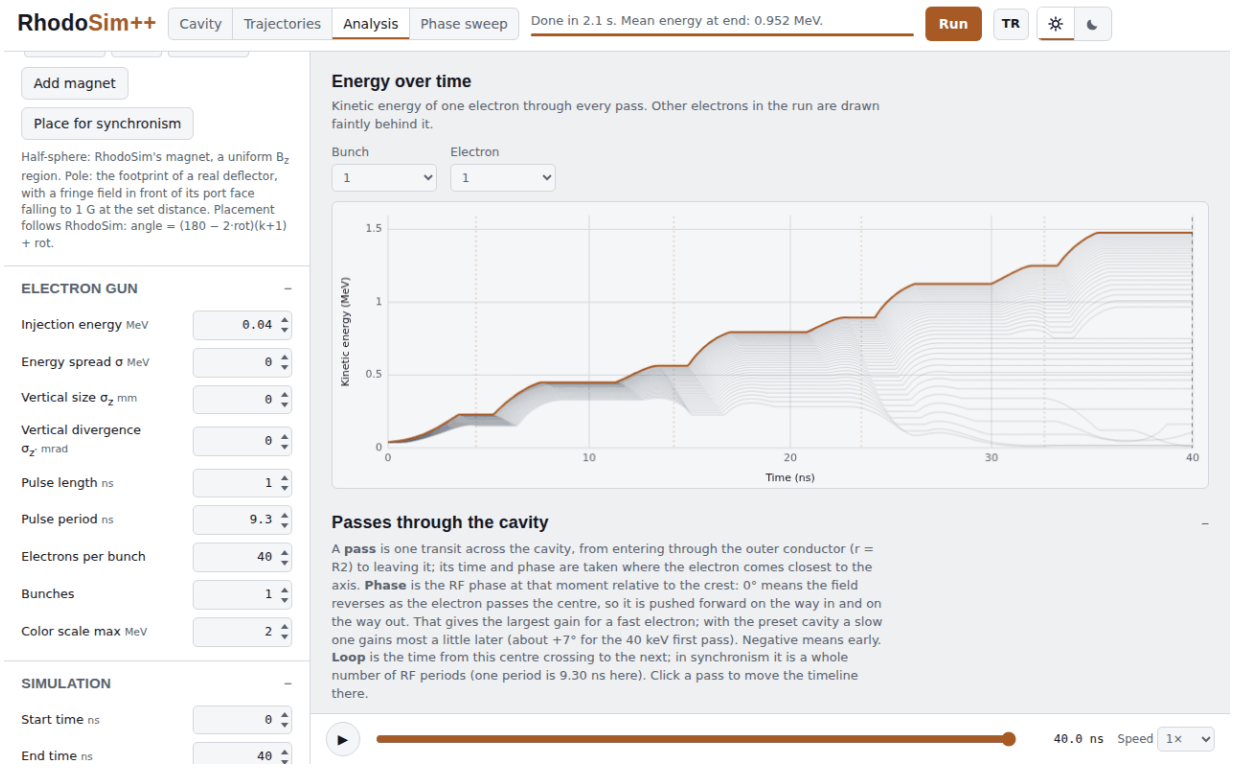


Figure 5: The Analysis tab: the energy over time and the selected electron's passes through the cavity; the energy distribution and the bunch's summary follow further down.

Pass	Electrons	Energy after (MeV)	Phase (°)	Loop (RF periods)
1	40 of 40	0.4112 ± 0.0357	28.1 ± 12.2	1.077
2	40 of 40	0.6259 ± 0.1456	55.8 ± 18.9	1.025
3	40 of 40	0.7765 ± 0.3258	63.5 ± 29.5	1.022
4	25 of 40	1.2849 ± 0.1498	37.9 ± 10.8	–

Figure 6: The bunch, pass by pass. A run with vertical motion adds the vertical beam size and divergence at each pass's centre.

When the run has vertical motion, the summary adds the vertical size (σ height, mm) and divergence (σ angle, mrad) at each pass's centre, and Vertical phase space plots every electron's height against its angle for a chosen pass, with the losses by cause.



Figure 7: The vertical phase space at the first pass. The upward tilt is a diverging beam: higher electrons are moving upwards.

5 Phase sweep

The Phase sweep tab runs the machine over a range of RF phase lags and plots a chosen quantity at a chosen time: Mean energy, Energy spread σ_E, Position spread σ_R, or Transmission, the share of electrons not lost. The chart marks the phase with the highest mean energy, the one with the smallest energy spread above 80 % of that, and the current phase lag (dashed). Click a point to adopt its phase lag.

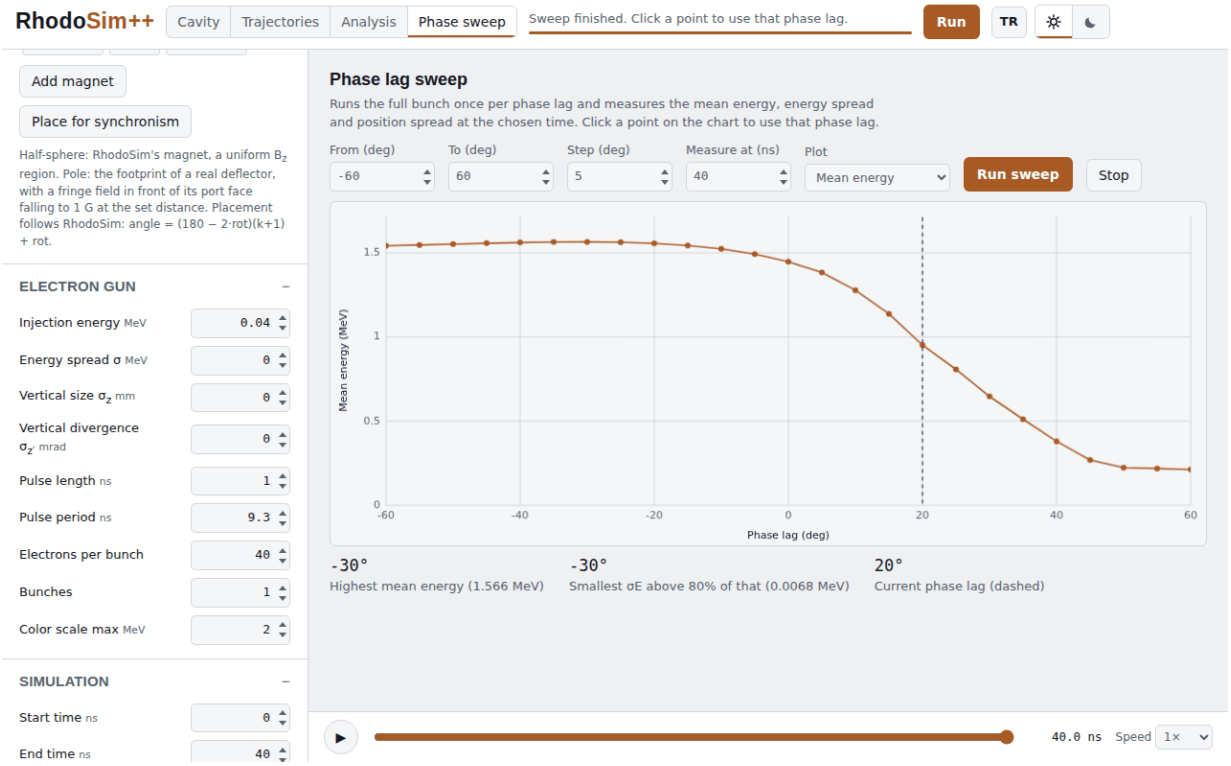


Figure 8: A phase sweep from $-60°$ to $60°$ in $5°$ steps.

6 The Cavity tab

The Cavity tab designs the RF cavity. A rhodotron cavity is a coaxial line shorted at both ends, a body of revolution; the tab solves its resonant mode in the r–z half-section and shows the fields, the cavity's figures of merit, and their accuracy.

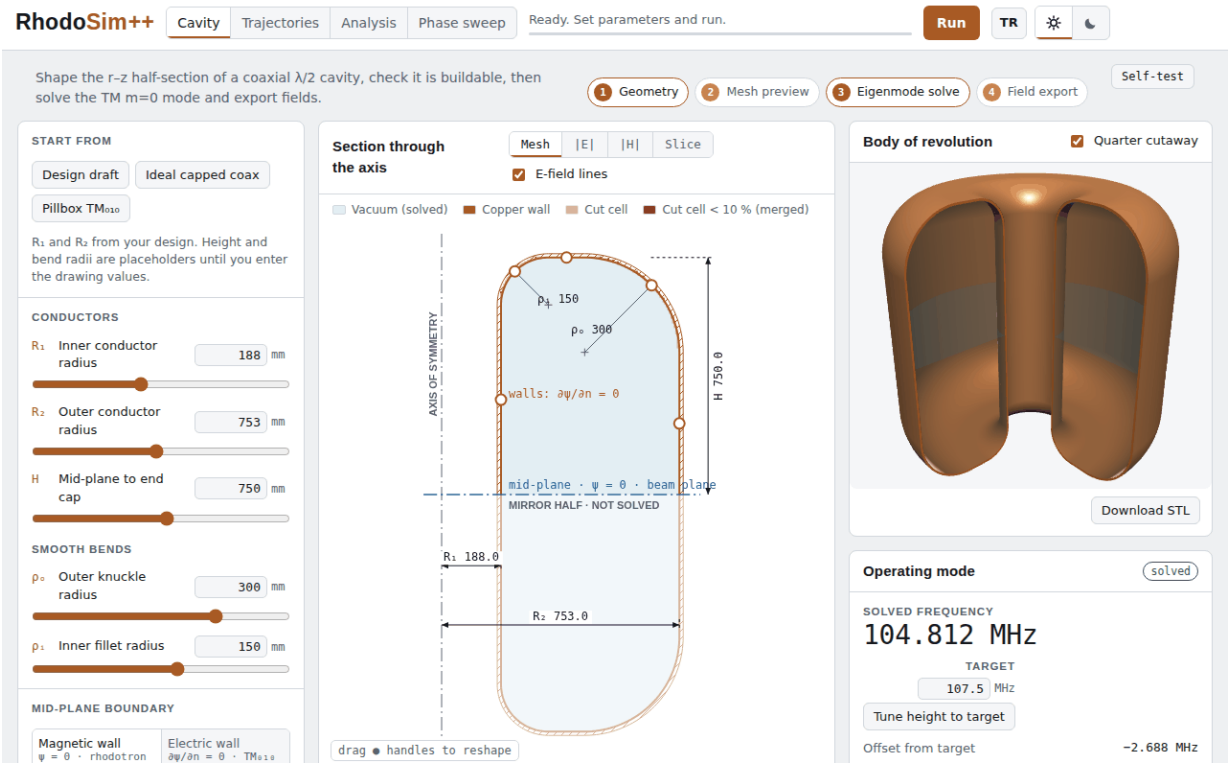


Figure 9: The Cavity tab: the parameters, the section through the axis, the 3D view, and the readouts.

Parameters. Start from offers example cavities. The section is set by the inner and outer radii R_1 and R_2, the half-height H, the rounding of the outer corner (the knuckle) and of the inner one (the fillet), the wall thickness, and the mesh's cell size h. The Mid-plane boundary condition is a magnetic wall (the rhodotron's mode) or an electric wall. The mode is solved automatically after a change (Solve eigenmode); the status shows solved, solving, or out of date. A target frequency and Tune height to target adjust H until the cavity resonates at the target.

The section. Mesh, |E| and |H| show the mesh or the field magnitudes on the section; Slice shows a horizontal cut through the cavity.

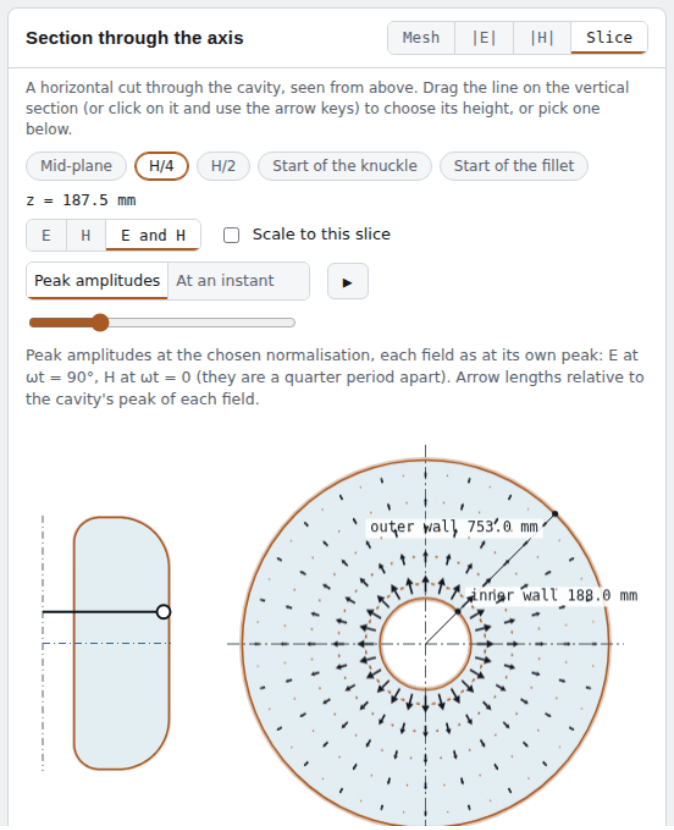


Figure 10: The slice view at a quarter of the height. Radial arrows are E_r, tangential ones H_ϕ, and the wash is E_z.

In the slice view, drag the height line on the small section (or use the arrow keys), or pick a height: Mid-plane, H/4, H/2, Start of the knuckle, Start of the fillet. E, H or E and H choose the arrows; Scale to this slice rescales them to the slice's own peak. Peak amplitudes shows each field at its own peak (they are a quarter period apart); At an instant shows both at a phase ωt set by the slider, and the play button animates one RF cycle.

Readouts. The frequency, the copper Q_0, the gap voltage, the stored energy, the wall loss and the peak surface field, with the mesh's size; Mid-plane field plots E_r along the mid-plane. The 3D view shows the cavity's copper body (a quarter is cut away); Download STL saves the full body as an STL file in millimetres.

Mesh convergence. Run convergence study solves the cavity at six cell sizes and extrapolates each quantity to an infinitely fine mesh. The table gives each quantity at your mesh, its extrapolated value, the estimated error at your mesh, and the observed order of convergence.

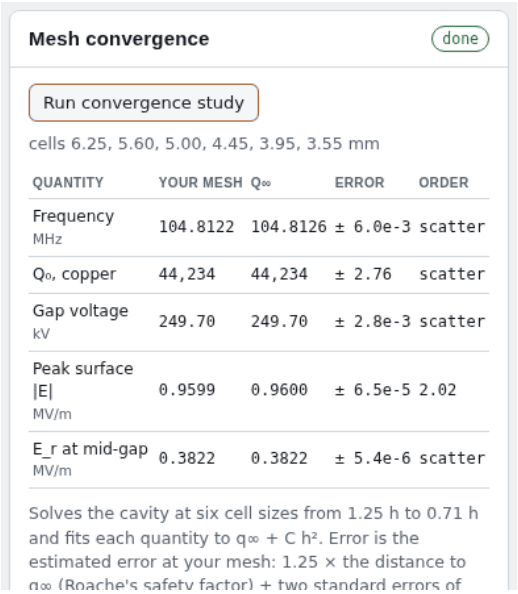


Figure 11: A convergence study. The error column is the uncertainty to quote with your mesh's values.

Field export. Export fields writes the solved field on a Cartesian grid for RhodoSim, as text or binary, in CST's nine-column format, or the native r–z grid; Copy config.ini lines gives the matching RhodoSim settings.

7 Designing a machine

RhodoSim++ has two uses. For learning and simulation, start in Trajectories and explore. For design, the usual path is frequency, cavity, magnets, beam, iterating. The tabs keep their own values and never change each other on their own; values move only when you ask:

- Design an RF cavity for this machine (Trajectories) opens the Cavity tab with the machine's radii and RF frequency as the target.
- Use this cavity in Trajectories (Cavity) sends the radii, the resonant frequency and the solved field to the beam, and selects the solved field. Send field map to Trajectories sends the field alone.

When the tabs disagree, warnings say so. Amber warnings point out differences: the tabs' radii or frequencies differ, the cavity does not resonate at the RF frequency, or the magnets were placed for another R_2 (Keep them accepts them). Red warnings mark a setup that is not physical: a field map for another cavity, a magnet inside the outer conductor, or two magnets whose fields overlap. A non-physical setup runs only after confirmation, and its results are labelled.

8 The physics model

Motion. Electrons obey the relativistic Lorentz force,

$$\frac{d\mathbf{p}}{dt} = q\,(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \tag{1}$$

in RhodoSim's units (ns, m, MV/m, T, MeV). An electron starts moving at the first time step after it is fired, as in C++ RhodoSim.

RF field. The analytic field is the $\lambda/2$ coaxial mode,

$$E_r = E_{\max}\frac{R_1}{r}\cos(kz)\sin(\omega t+\varphi), \qquad B_\phi = -\frac{E_{\max}}{c}\frac{R_1}{r}\sin(kz)\cos(\omega t+\varphi), \qquad k=\frac{\omega}{c}, \tag{2}$$

between the conductors; in the mid-plane ($z=0$) only E_r remains. The solved field replaces it with the Cavity tab's map of E_r, E_z and B_ϕ over the whole cavity.

Magnets. A half-sphere magnet has a uniform B_z inside a ball of radius R, beyond a face plane $R/5$ in front of its centre. A pole magnet follows a deflector drawing; in front of its face the field rises as $B/(1+e^{-d/\lambda})$, with λ set so that $|B| = 1\,\mathrm{G}$ at the chosen distance. Off the mid-plane the fringe carries the components Maxwell's equations require, $B_d = z\,\partial B_z/\partial d$ to first order, which focus the beam vertically at the face; the edge focusing follows Brown's formula of accelerator optics.

Integrators. The Boris scheme (second order, energy-conserving in magnetic fields), the classical fourth-order Runge–Kutta scheme, and RhodoSim's own scheme, which reproduces C++ RhodoSim's results.

The cavity solver. The Cavity tab solves the TM mode ($m=0$) for $\psi = rH_\phi$,

$$\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial\psi}{\partial r}\right)+\frac{\partial}{\partial z}\left(\frac{1}{r}\frac{\partial\psi}{\partial z}\right)+k^2\frac{\psi}{r}=0, \tag{3}$$

by finite volumes with cut cells at curved walls (cells less than 10 % open are merged). The fields follow from ψ: $H_\phi = \psi/r$, $E_r = -\frac{1}{\omega\varepsilon_0 r}\partial\psi/\partial z$, $E_z = \frac{1}{\omega\varepsilon_0 r}\partial\psi/\partial r$. At a 5 mm mesh the frequency agrees with exact solutions to a few parts per million.

9 Files

config.ini. RhodoSim's configuration file, one `key=value` per line; lines starting with # are comments, and keys are not case-sensitive. Edit config.ini shows the current settings in this form, applies edited or pasted text, and saves the file. RhodoSim++ reads and writes C++ RhodoSim's keys:

Key	Unit	Meaning
`emax`	MV/m	the peak field at the inner conductor (Peak E at R1)
`freq`	MHz	the RF frequency
`phaselag`	degrees	the RF phase lag
`r1`, `r2`	m	the inner and outer conductor radii
`magrotation`	degrees	the magnets' entry rotation
`addmagnet`	T, m, m	one magnet per line, written `B , R , <dist>`: its field, its radius and its distance from the centre; the lines in a file replace the magnet list
`einmean`	MeV	the injection energy (`ein` is read too)
`einstd`	MeV	the energy spread σ
`targeten`	MeV	the top of the energy colour scale (Color scale max)
`guntime`, `gunperiod`	ns	the gun's pulse length and pulse period
`enum`		electrons per bunch
`bunchnum`		the number of bunches
`starttime`, `endtime`	ns	the start and end of the run
`dt`	ns	the time step (it must be positive)
`multh`, `thcount`		C++ RhodoSim's threading: it moves electrons only in its multi-threaded path, so RhodoSim++ writes `multh=1` and `thcount=8` for it, and ignores both when reading

It adds its own keys, which C++ RhodoSim ignores:

Key	Unit	Meaning
`integrator`		`boris`, `rk4` or `rhodosim`
`magmodel`		`sphere` (half-sphere) or `pole`
`fringe1G`	m	pole magnets: the distance at which the fringe falls to 1 G
`fieldmodel`		`coax` (analytic) or `solved` (the Cavity tab's field map)
`zsig`, `zpsig`	mm, mrad	the gun's vertical size and divergence
`portdia`	mm	the port diameter
`polegap`	m	the pole gap

A file must contain at least one known key; unknown keys are ignored, so files written by C++ RhodoSim load as they are.

Field maps and STL. The Cavity tab exports field maps for RhodoSim (text, read by RhodoSim's imported-field model, or binary), in CST's nine-column format, or as the native r–z grid, and the cavity's body as STL.

10 Checking the results

Both halves of RhodoSim++ carry a self-test that you can run in your own browser. Each check compares the program with an *independent* oracle (an exact solution, a conservation law, a separate calculation, or recorded results) and shows the measured and expected values, the error and the tolerance.

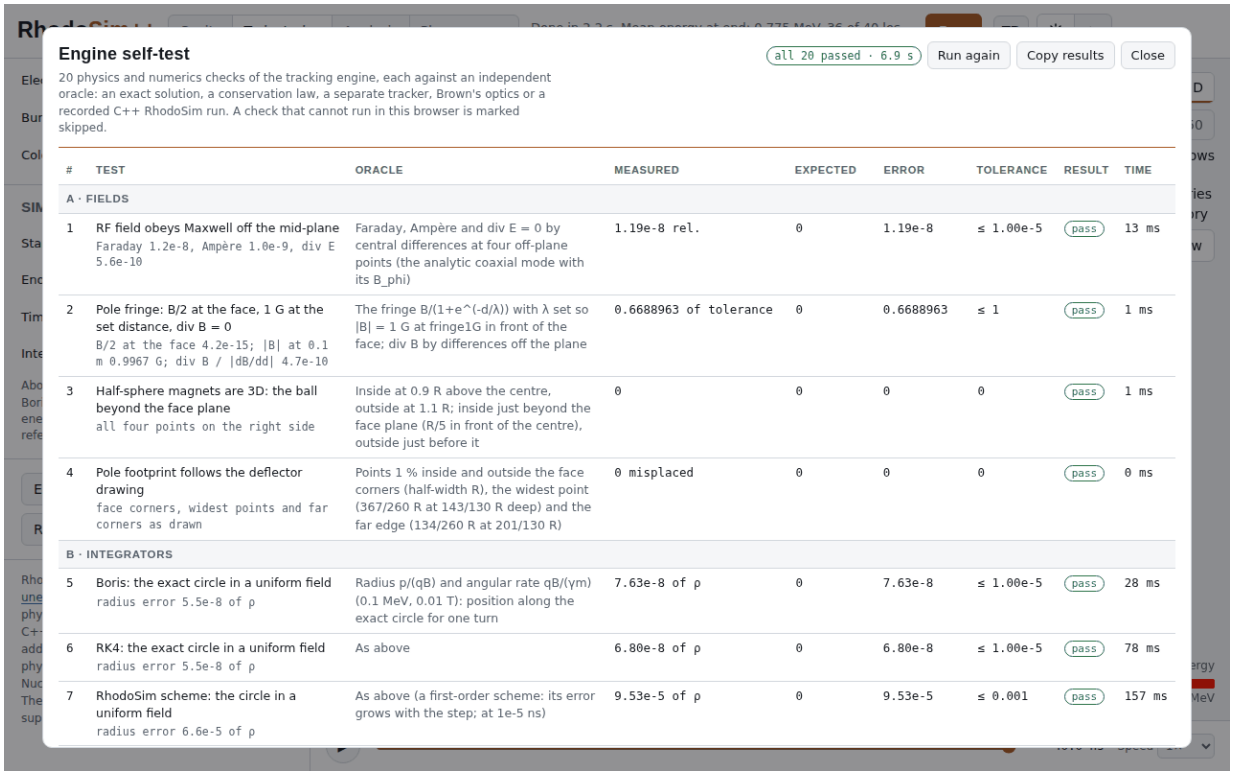


Figure 12: The engine's self-test (Self-test in the Simulation section).

- **The engine's self-test** (Self-test in the Simulation section): 20 checks in six groups: fields (Maxwell's equations, the magnet shapes), integrators (exact motion in a uniform field, energy conservation, orders of convergence), acceleration (an independent tracker), 3D motion (symmetry, edge focusing),

losses (exact times), and consistency (Web Workers against the main thread, `config.ini`, and C++ RhodoSim's recorded results).

- **The cavity solver's self-test** (Self-test in the Cavity tab): 49 checks of the geometry, the mesh, the solver against exact modes, perturbation theory, the field identities, the export and mesh convergence.

A check that cannot run in your browser is marked *skipped* rather than passed. Behind both stands a larger test suite run during development, including comparisons with C++ RhodoSim built from its source.